ENGR 4399 · PROJECT 2 · WIRING GUIDE

Mesh Barista

How to wire the three ESP32 nodes, check each part from the dashboard, and fix the common faults.

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Dashboard join the nodes' Wi-Fi, then open http://barista.io

BEFORE YOU START

Parts and ground rules

Build and test one node at a time. Unplug USB whenever you move a wire.

Node 1 · pump station

- [ ] ESP32 DevKit
- [ ] 3 relay modules
- [ ] 3 pumps and tubing
- [ ] 3 water sensors
- [ ] 1 buzzer (3-pin)
- [ ] Pump power supply
- [ ] 3 diodes (1N4007)
- [ ] 3 resistors, 10 kΩ (recommended)
- [ ] 3 reservoir cups

Node 2 · control center

- [ ] ESP32 DevKit
- [ ] Joystick module
- [ ] OLED, 0.96 in, 4-pin
- [ ] 1 buzzer (3-pin)
- [ ] USB cable to a laptop

Node 3 · output

- [ ] ESP32 DevKit
- [ ] 1 water sensor
- [ ] OLED, 0.96 in, 4-pin
- [ ] RGB LED, 4-pin (V G B R)
- [ ] Output cup

Four rules for every node

Signal parts run on 3V3

Water sensors, the joystick and the RGB LED go on the `3V3` pin, even where the part says `+5V`. More than 3.3 V into an ESP32 input can damage it.

Only relays use 5 V

Relay modules take their power from the `VIN` (or `5V`) pin.

All grounds join

Every part's GND goes to an ESP32 `GND` pin. On Node 1 the pump supply's negative goes there too.

Pumps have their own supply

Never power a pump from an ESP32 pin. The relays switch the pump supply.

Reading the diagrams

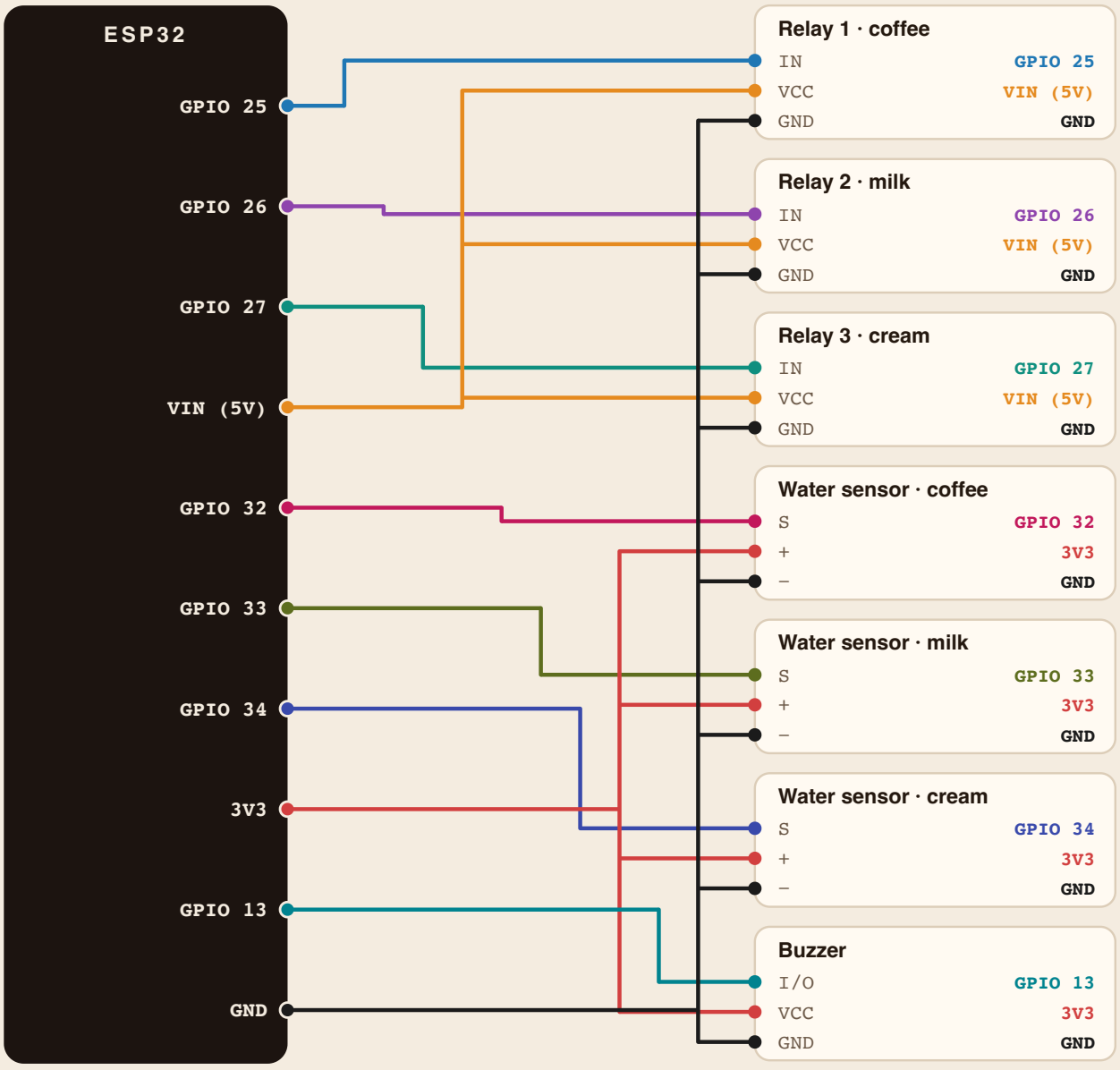
Each line is one jumper wire from the ESP32 pin on the left to the part pin on the right. Pin names are GPIO numbers; most boards print them as `D25`, `D32` and so on. Run 3V3 and GND to the breadboard's power rails first, so every part takes power from the rails.

3V3 power · 5 V power (VIN) · Ground · Signal wires

Pump station: control wiring

The ESP32 side: relays, reservoir sensors and the buzzer. No pump power yet.

Node 1 connections



Relay jumper

If a relay module has an H/L jumper, set it to **H** (active-HIGH).

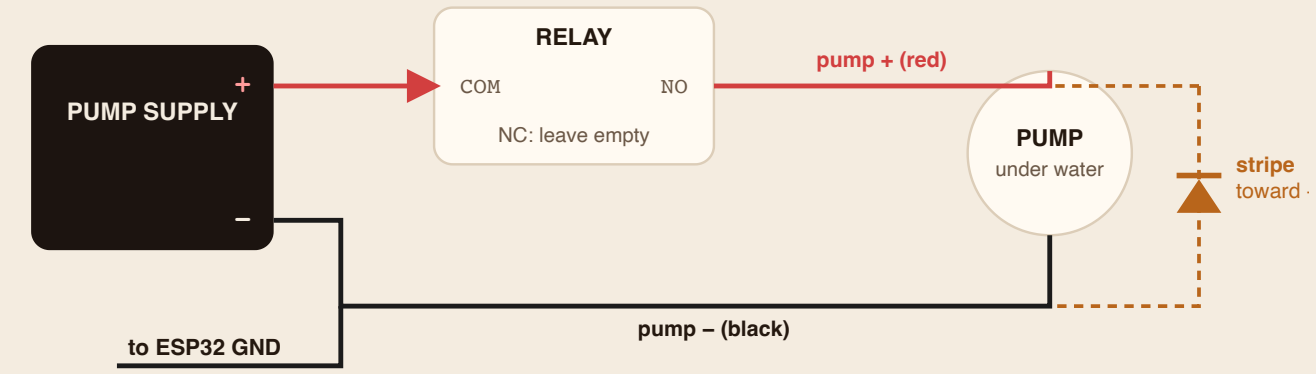
Pull-down resistors

One 10 kΩ from each of GPIO 25, 26 and 27 to GND holds the relays off while the ESP32 resets.

Pump station: pumps and water

The pump side never touches an ESP32 pin except ground. Repeat this circuit for each of the three relays.

One pump circuit



1. Pump supply **+** to the relay's **COM** terminal.
2. The relay's **NO** terminal to the pump's **+** wire (usually red). Leave **NC** empty.
3. The pump's **–** wire (usually black) to pump supply **–**.
4. Pump supply **–** to an ESP32 **GND** pin. One wire covers all three pumps.
5. A diode across each pump's two wires, with the **striped end on the pump's + wire**.

Check the diode direction

A backwards diode shorts the supply the moment the relay closes. The relay clicks and the pump never runs.

Why NO and not NC

On NO, a pump only runs while the ESP32 is actively holding its relay on. A reset or power loss leaves every pump off.

Water and tubing

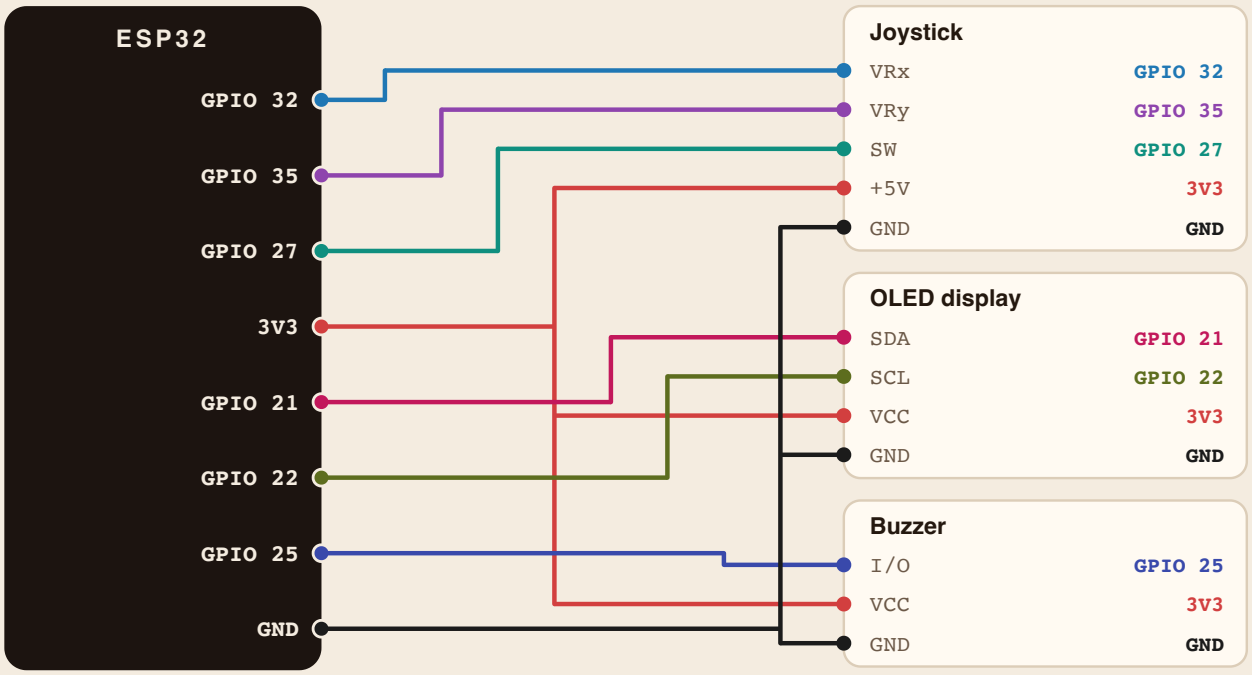
- [ ] Each pump sits at the bottom of its reservoir. A submersible pump must be under water to pump.
- [ ] Each water sensor is clipped inside its reservoir, strips pointing down, electronics at the top kept dry.
- [ ] The three outlet tubes reach the cup on Node 3.
- [ ] A tray sits under everything to catch spills.

NODE 2

Control center

The joystick picks the drink, the OLED shows the menu, and the buzzer sounds the alarm.

Node 2 connections



Joystick power

The joystick's power pin is labeled `+5V`, but it must go to **3V3**.

Hands off at power-up

Node 2 measures where the joystick rests when it starts. Leave the stick alone for the first second.

Using the joystick

MOVE	WHAT IT DOES
Left / right	Choose coffee, milk or cream
Up / down	Change the amount in 10 mL steps
Click	Pour. In SAFE, a click acknowledges instead

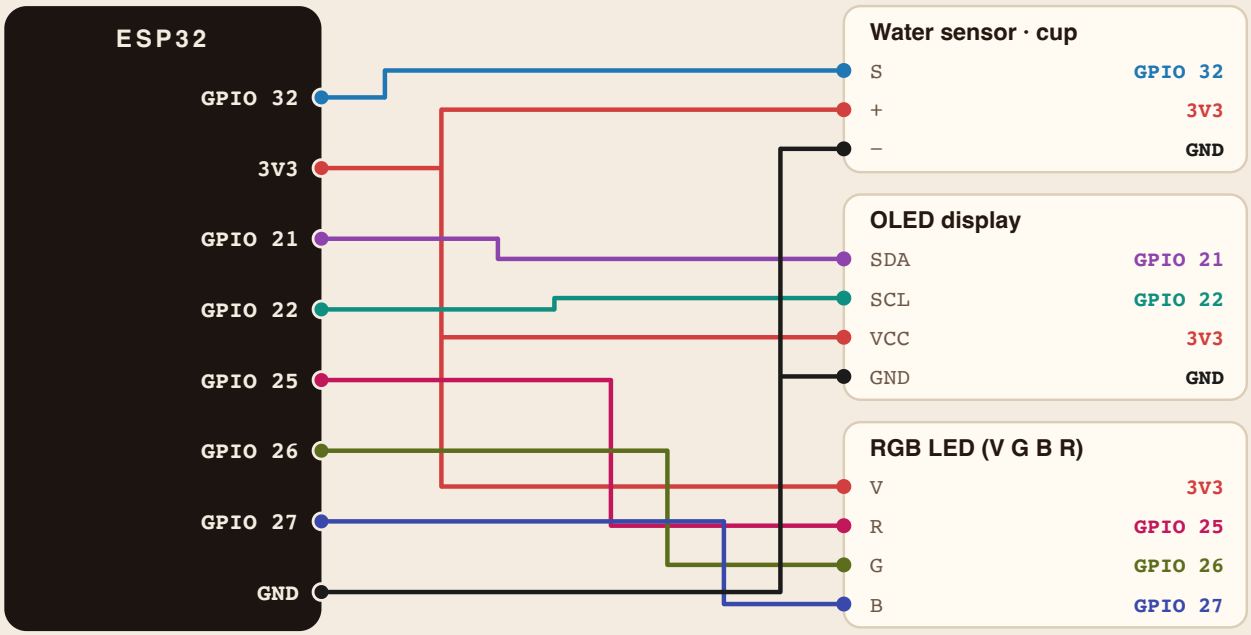
Keep this node on USB to a laptop if you want the text dashboard: Serial Monitor at 115200 baud.

NODE 3

Output

The cup sensor's reading is what stops the pump. The LED shows which pump is running.

Node 3 connections



RGB LED

The LED is common-anode: V is the shared positive. If yours is a bare LED with no small board, put a 220 Ω resistor in line with each of R, G and B.

Cup sensor

Clip the sensor inside the cup so the water reaches the strips as it fills. The top of the strips is the full line.

What the LED means

COLOR	MEANING
Green	Ready
Orange	Coffee pouring
White	Milk pouring
Yellow	Cream pouring
Flashing red	SAFE or DEGRADED: check the dashboard

POWER-UP

Checklist

Work down the list in order. Join the Wi-Fi the nodes create and open http://barista.io (or http://192.168.4.1).

- [ ] **Each node alone on USB.** The OLED lights up on Nodes 2 and 3. Node 1 prints the dashboard address on Serial.
- [ ] **All three on.** The Live tab shows three nodes ALIVE. The system starts in SAFE: press Acknowledge, or click the joystick.
- [ ] **Buzzers.** Test tab: Beep Node 1 buzzer, then Beep Node 2 buzzer. Each beeps for one second.
- [ ] **LED.** Cycle Node 3 LED shows red, then green, then blue.
- [ ] **Joystick.** The Inputs card shows X and Y near 1900 at rest, moving toward 0 and 4095 when pushed.
- [ ] **Water sensors.** Note each raw value dry and at the full mark. Put them in `LEVEL_RAW_DRY` / `LEVEL_RAW_FULL` on Node 1 and `CUP_RAW_DRY` / `CUP_RAW_FULL` on Node 3, then flash again.
- [ ] **Pumps, with the pump supply on.** Coffee pump 10 mL, then milk, then cream. Each relay clicks and its pump runs briefly.
- [ ] **Flow rate.** Pour 200 mL into a measuring cup and time it. Put mL divided by seconds into `FLOW_ML_PER_S` on Node 1, for each pump.
- [ ] **Fail-safe.** Unplug Node 3 during a pour. The pump stops, both buzzers sound, and the dashboard names the output zone.

> **Calibration record**
> Write your numbers here so they survive the next flash.

SENSOR	RAW, DRY	RAW, FULL	PUMP ML PER SECOND
Coffee reservoir			
Milk reservoir			
Cream reservoir			
Cup			not used

> **Phone tips**
> Turn mobile data off so the name barista.io is looked up on the nodes' Wi-Fi, and type the `http://` part.

TROUBLESHOOTING

If something does not work

Find the symptom on the left. Causes are listed most likely first.

SYMPTOM	LIKELY CAUSE
Relay clicks, pump does not run	Diode backwards. Pump on NC instead of NO. Pump supply off or too weak. Pump not under water. Test the pump by touching its wires straight to the supply.
Relay clicks on at boot and off when a pour starts	Active-LOW relay module. Set `RELAY_ON` to `LOW` on Node 1.
Relay does not click at all	Relay VCC is not on VIN (5V), or the module will not switch from a 3.3 V signal.
Buzzer sounds all the time	Active-HIGH buzzer module. Set `BUZZER_ON` to `HIGH` in that node's sketch.
Level stays at 0 % or 100 %	Sensor on the wrong pin, or the raw dry and full values are not calibrated yet.
OLED is blank	SDA and SCL swapped, or a display that is not at address 0x3C.
Menu will not move	The joystick was held during power-up, or VRx is not reaching GPIO 32. Check Joystick X on the Test tab.
LED colors are wrong	Two of the R, G, B wires are swapped. Run Cycle Node 3 LED to see which.
A node shows LOST	That board is unpowered, or was flashed with a different `MESH_SSID` or `MESH_CHANNEL`.
barista.io will not load	Not joined to the nodes' Wi-Fi, mobile data is on, or `http://` was left off. Use http://192.168.4.1 instead.
Stuck in SAFE	Acknowledge is refused while Node 3 is LOST. Bring Node 3 back, then acknowledge.